IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR

AIM:

To implement classless IP subnetting using CISCO PACKET TRACER and verify connectivity between different subnets.

PROCEDURE:

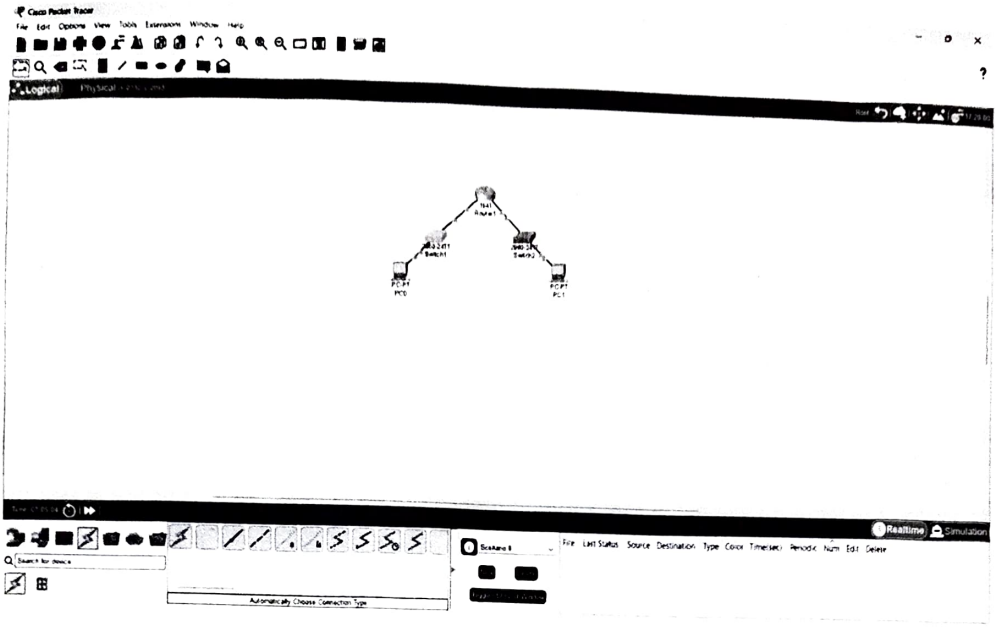
1. Open Cisco packet Tracer and create a new network topology.
2. Add routers, switches and PCs to workspace
3. Connect the devices using appropriate cables.
4. Divide the network 192.168.0.1/024 into smaller subnets using 127 subnet mask.
5. Configure the IP address and subnet mask for the router interfaces.
6. Configure the PC's with suitable IP addresses, subnet masks and default gateways.
7. Configure the Switch ports and router interfaces
8. Use the PING command to test connectivity between the devices.
9. Verify that the Packets are successfully transmitted between the connected networks.

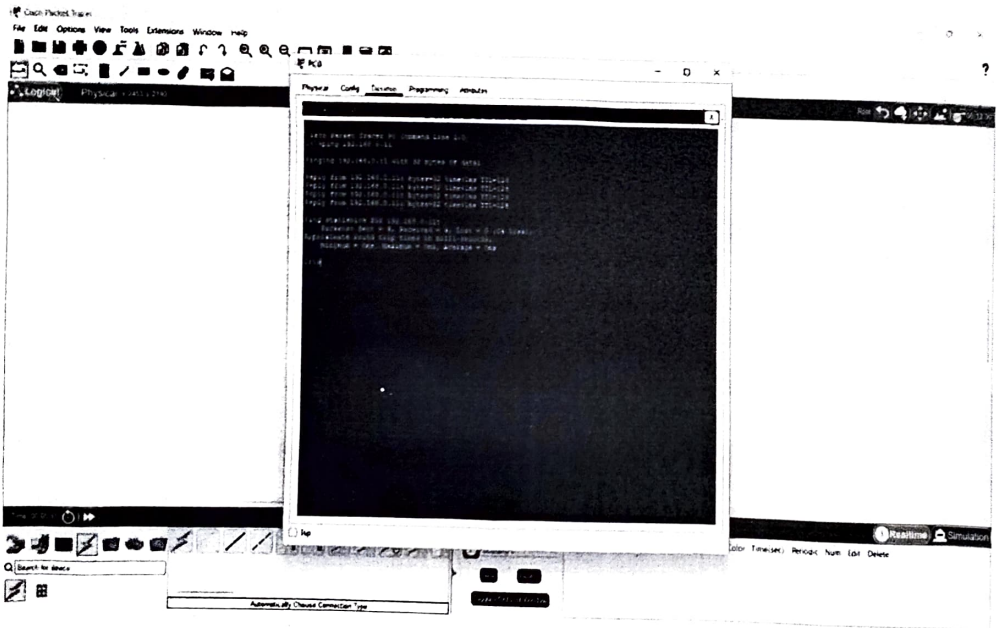
IP addressing:

Network	Subnet mask	Router IP	PC IP Range
192.168.1.0/27	255.255.255.224	192.168.1.1	192.168.1.11.15
192.168.2.0/27	255.255.255.224	192.168.2.1	192.168.2.11.15
192.168.3.0/27	255.255.255.224	192.168.3.1	192.168.3.11.15
192.168.4.0/27	255.255.255.224	192.168.4.1	192.168.4.11.15

Result:

The classless IP subnetting was successfully implemented using CISCO PACKET TRACER.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Realtime
Simulation
Automatically Choose Connection Type


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
PC0
Physical Config Desktop Programming Attributes
Logical
Realtime
Simulation
Automatically Choose Connection Type